

The Battle Of The Blue

Your new DVD-Writer will soon be obsolete. But which blue-laser technology will replace it?

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We don't need to tell you that technology moves at a fast pace, do we? But we have here an indicator of just how fast: in September of 2004, in *Space: The Final Frontier*, we talked about the DVD format wars. Back then, we just mentioned Blu-ray and HD-DVD in passing. Just about a year later, the DVD format wars have been all but forgotten. Blu-ray and HD-DVD are already upon us, and we need to talk about them in much more detail than we did last year.

For the uninitiated, Blu-ray and HD-DVD are the next generation of optical storage, though there do exist several competing formats as well. Blu-ray offers about 25 GB of storage on a single layer, but a Blu-ray Disc (BD) can and will host more than one layer—hence a typical dual-layer

BD will be capable of holding up to 50 GB. HD-DVD offers about 15 GB on a single layer. Dual and triple-layer discs of this format have been manufactured.

Again, for those who came in late, there's a format war of unprecedented proportions taking place. The Internet is abuzz with discussions, news, views, and speculation on which format—which camp, so to speak—will emerge the winner and become the standard, whether there will be a peaceable co-existence, and a host of other things. What follows is a W5H of sorts on Blu-ray and HD-DVD, followed by the much-asked question, "Which format will win?"

So What Are Blu-Ray And HD-DVD?

The Blu-ray standard was developed by a group of companies called the Blu-ray Disc Association (BDA). Blu-ray gets its name from the "blue-violet" laser used to write the data to disc. (Different laser wavelengths have different colours.) The blue-violet laser has a wavelength much shorter than that used to write conventional DVDs, which happens to be red. The shorter wavelength makes it possible to read and write smaller "pits," meaning that data can be packed more densely.

Ones and zeroes are stored on optical media—CDs and DVDs included—in the form of



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